

GC Analysis Application where  
LabVantage has all the Data



# GC Analysis Application Objects



- Material Standard
- Sample Metadata
- Raw GC Run Data (from Instrument)
- Instrument Calibration Data
- GC Run Analysis Data (added by Analyst)
- Manifest
  - Enumerates the above objects needed to do GC analysis for one Sample or one GC Run (not sure which yet) including the objects needed to produce what is now called the History Report
  - Lists object types, keys and dates last changed
- GC Analysis Work Queue (of a particular type)
- Association of User to GC Analysis Work Queue(s)



# GC Analysis Work Queue Types



- One per Lab (Non-Coded)
  - Calibration
  - Easy Non-Coded Samples (or GC Runs)
  - Hard Non-Coded Samples (or GC Runs)  
10% to be Reviewed (Valhalla does these)
- Valhalla Only (Coded)
  - Samples (or GC Runs)



# Data Stores



- LabVantage
  - Material Standards
  - Sample Metadata
  - Raw GC Run Data (from Instrument)
  - Instrument Calibration Data
  - GC Run Analysis Data (from GC Analysis)
  - Everything Else
- GC Analysis App
  - Work Queues
  - Cache of Objects which support Work in Queues
  - Users and their assigned Work Queues (could be stored in MyIDm)

# API's



- LabVantage
  - Provides
    - Material Standard
    - Sample Metadata
    - Raw GC Run Data
    - Instrument Calibration
    - GC Run Analysis Data
    - Manifest
  - Accepts
    - GC Run Analysis Data
    - Instrument Calibration
- GC Analysis App
  - Accepts
    - Work Queue Entry (from LabVantage)

# Events



- LabVantage
    - Receive Raw GC Run Data (from Instrument)
    - Receive GC Run Analysis Data (created by Analyst using GC Analysis App)
    - Receive Instrument Calibration (created by Analyst using GC Analysis App)
  - GC Analysis App
    - Receive Work Queue Entry (from LabVantage)
    - User Logs In
    - User Selects a Sample or GC Run for Analysis
    - User Requests the History [Report] related to the GC Run or Sample being analyzed
    - User “Locks” GC Run
    - Every so often (a recurring timer)
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# Receive Work Queue Entry Event



- Add Entry to Queue
- Request Manifest of Objects needed to perform GC Analysis and provide History
- See if objects are cached:
  - If old object is cached, replace with new object from LabVantage
  - If object is not cached, get it from LabVantage



# User Logs In Event



- Figure out which Work Queues he is supposed to work on
- Show him and let him search the work in his Work Queues





# User Selects a Sample Event



- Request Manifest of Objects needed to perform GC Analysis and provide History for the Sample (or GC Run, not sure which is getting analyzed)
- See if objects for selection are cached (even though they should already be cached):
  - If old object is cached, replace with new object from LabVantage
  - If object is not cached, get it from LabVantage
- Open the Chromatogram using cached objects



# User requests the History Event



- Request Manifest of Objects needed to perform GC Analysis and provide History for the Sample (or GC Run, not sure which is getting analyzed)
- See if objects for selection are cached (even though they should already be cached):
  - If old object is cached, replace with new object from LabVantage
  - If object is not cached, get it from LabVantage
- Display the History using cached objects



# User “Locks” GC Run Event



- If it is a Calibration Run, send Instrument Calibration to LabVantage
- If it is a Normal Run, send GC Run Analysis Data to LabVantage

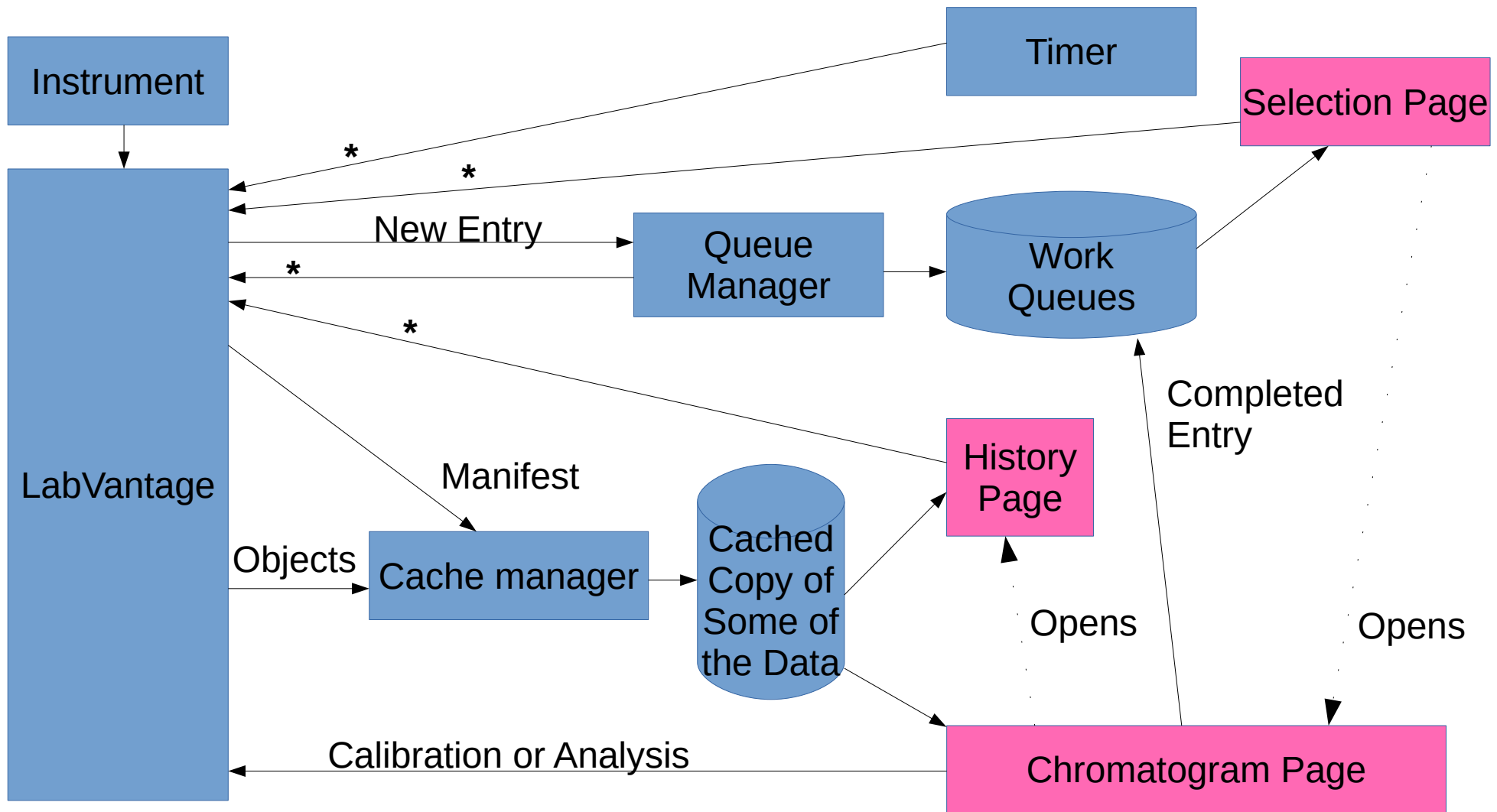


# Every so often Event



- Refresh Cache of Objects which support the Queued Work by doing the following for each Entry in each Work Queue
  - Request Manifest of Objects needed to perform GC Analysis and provide History for the Sample (or GC Run, not sure which is getting analyzed)
  - See if objects for selection are cached (even though they should already be cached):
    - If old object is cached, replace with new object from LabVantage
    - If object is not cached, get it from LabVantage

# Implementation



**\* Requests for Current Manifest**